Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 5.1 · A table

A survey covers 100 users of two transport services A and B. Of these, 40 use A, 35 use B and 15 use both. One of the 100 users is chosen at random; all have the same probability of being chosen. The events A and B respectively mean "the user uses the service A" and "the user uses service B".

1. Construct and complete a two-way table: use of A or not in rows, use of B or not in columns. Add the totals.
2. Calculate $P_A(B)$ and $P_B(A)$. For each calculation, state the reference population used in the denominator.
3. Interpret both probabilities with two separate sentences.

Exercise 5.2 · Production

A company manufactures objects in two workshops A and B, which produce respectively $\frac{1}{4}$ and $\frac{3}{4}$ of the production. Among the items produced by A, 4 per cent are defective; among those produced by B, 2 per cent are. An item is chosen at random from the whole production. Let D the event "the item is defective".

1. Represent the situation with a weighted tree and state the meaning of each probability on the second level.
2. Calculate $P(D)$ and interpret the result as an overall defect rate.
3. We now know that the chosen object is defective. What is the probability that it was made in workshop A?

Exercise 5.3 · Independence to be verified

A fair die with six faces numbered from 1 to 6 is rolled. Consider the events A : "getting an even number" and B : "getting a number greater than or equal to 4".

1. Write out the outcome sets corresponding to A, B and $A \cap B$.
2. Determine whether the events A and B are independent. Justify using a numerical criterion.
3. Explain why they are not disjoint either.

Exercise 5.4 · Without replacement

An urn contains three red balls and two blue balls, indistinguishable to the touch. A first ball is drawn at random, then a second without replacing the first. At each draw, all the balls in the urn are equally likely.

1. Construct the weighted tree of the two draws, distinguishing the colour obtained on the first and on the second draw.
2. Calculate the probability of getting two red balls.
3. Calculate the probability of getting one ball of each colour. Explain why two paths of the tree must be taken into account.

Exercise 5.5 · Three trials

An experiment has a success with probability $\frac{1}{3}$ and a failure. It is repeated three times independently, under the same conditions.

1. Construct a three-level tree, or list the eight possible sequences of outcomes.
2. Calculate the probability of exactly one success. State the paths used.
3. Calculate the probability of getting at least two successes, distinguishing the cases of two and three successes.

Exercise 5.6 · Detector

A device checks objects of which 2 per cent are defective. It triggers a signal for 90 per cent of defective items and, wrongly, for 5 per cent of non-defective items. An item is picked at random. Let D the event "the item is defective" and S the event "the device triggers a signal".

1. Determine the proportion of actually defective items among the flagged items. Choose a representation and explain the reasoning in detail.
2. Give an exact value, then a percentage rounded to one decimal place.
3. A person immediately gives 90 per cent as the answer. Explain precisely which confusion of probabilities they make.

Exercise 5.7 · Sets and cardinality

We work in the set $E = \{1; 2; 3; 4\}$. Consider its subsets $A = \{1; 2\}$ and $B = \{2; 4\}$.

1. Write $A \cap B$, $A \cup B$ and the complement of A in E. Give the cardinality of each.
2. Write all pairs of $A \times B$, then give its cardinality.
3. Explain why the pairs (1; 2) and (2; 1) cannot be confused. Does the second belong to $A \times B$?

Exercise 5.8 · Delayed review

In a random experiment, two events A and B satisfy

$$P(A) = \frac{1}{2}, \qquad P_A(B) = \frac{1}{4}, \qquad P_{\overline{A}}(B) = \frac{3}{4}.$$

1. Calculate $P(B)$ using a tree or the law of total probability.
2. Calculate $P_B(A)$ by showing the common event used in the numerator.
3. The probabilities $P_A(B)$ and $P_B(A)$ equal here? Is this equality a general property ?

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 67 p. 336, questions 1, 2 and 3. Build a tree keeping the events D and N from the textbook. Note: N denotes a negative test.

CQFD, exercise 70 p. 336, questions a and b. Count the four equally likely outcomes of two rolls to refute the proposed equalities.